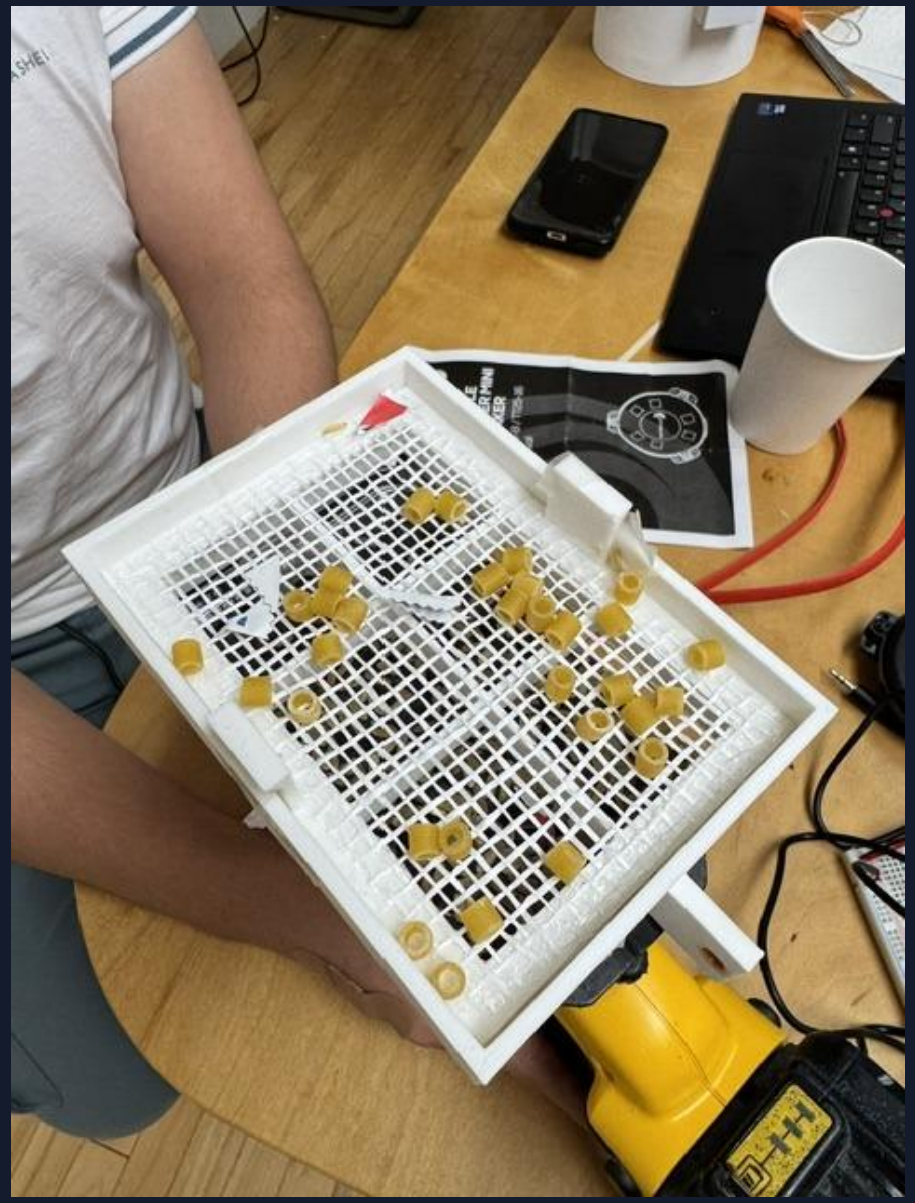
PHYSICAL AI FOR SCIENCE HACK · BIOKEA TRACK

BugSorter2.1: print, test, repeat

Problem: A sample of **thousands of tiny flies in ethanol,** tangled into a ball and mixed with leaves, twigs and dirt when poured into a BugSorter is not able to de-clump effectively with the current BioKeA set-up. The sample is hard to de-clump and impacts successful separation of biodiversity.

A loop for finding the right mesh and gate sizes ourselves. Every size is a number in a script: print it, test it on the bench, check it in a sim of the stack moving up and down.



Aufstin Filiko · Forum Shah
Studio 45, San Francisco · Oct 24–25, 2026

WHAT WE BUILT ON

Prototype 1

- **Demo rig:** a printed tray and sieve bolted to a handheld DeWalt 20V tool, for testing (without the tower)

- **Which size goes on each tray?** It depends on the sample, and on how specimens tumble and stand on end. The right size has to be found by testing.

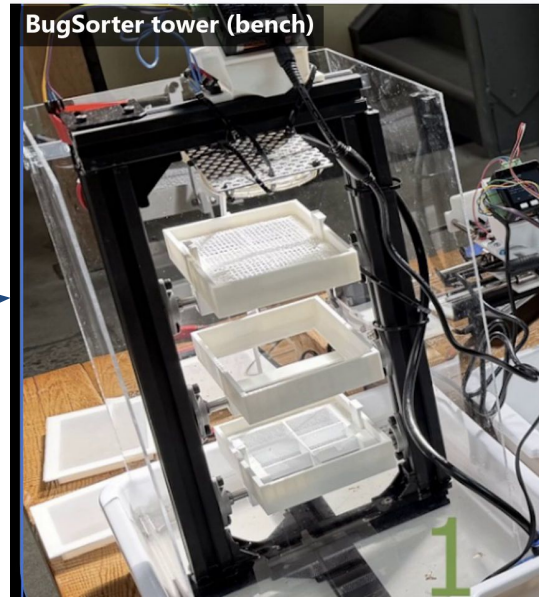
- **So we made sizes cheap to try.** Every mesh, gate and tray is a CadQuery script sized for BugSorter's holder (131.4 × 95.4 mm). Change a number, print, test.

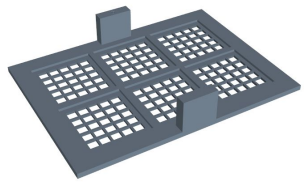


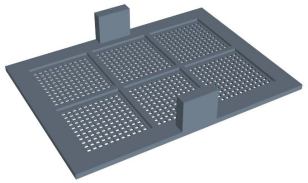
Demo rig: printed tray on a DeWalt 20V MAX tool

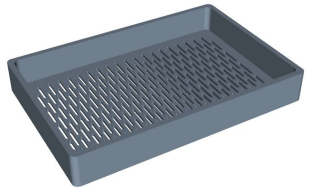
Comparison of current set up and the sim

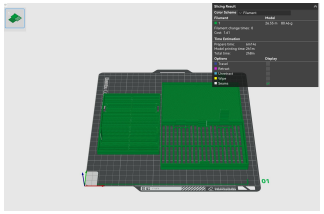


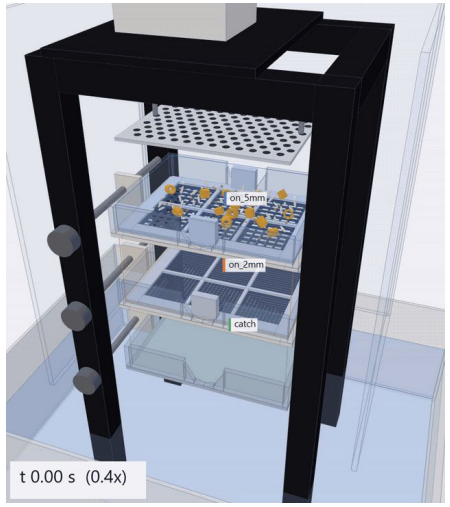












The sim runs a size before we print it. Here: the 5 mm tray, 2 mm tray and catch pan moving together.

Current clip: a 20 Hz, ±1.24 mm vertical stroke (assumed). Trays hung 65 mm apart, estimated from the photo. The frame, spray plate and tub are display only.

Initial sizes:

- > 6 mm
- 2–6 mm
- < 2 mm

DEMO · OUR RIG

▶ Full-res video

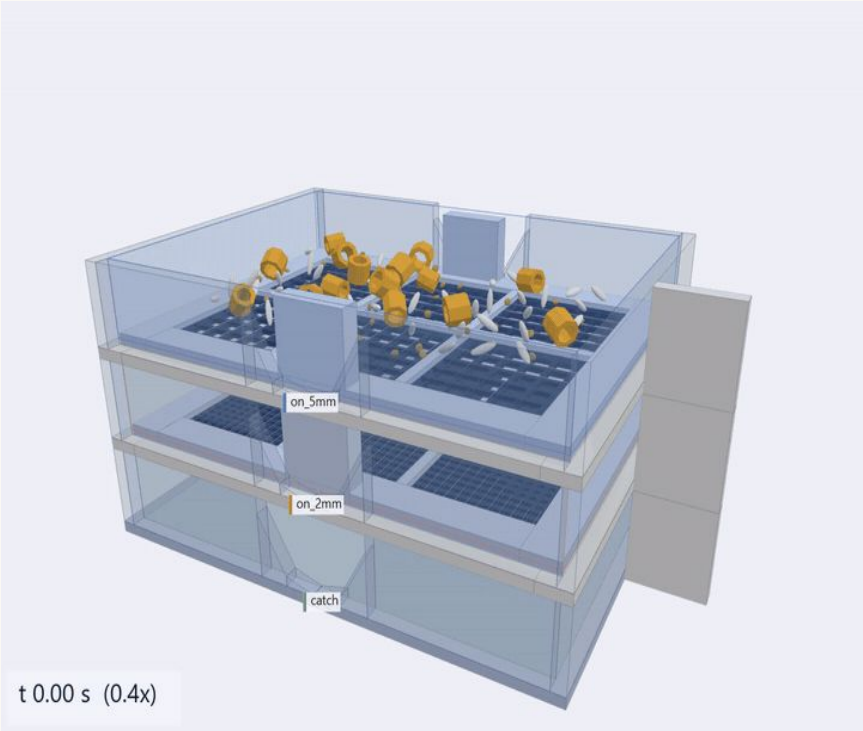
One tray on the tool, and the full stack it belongs in

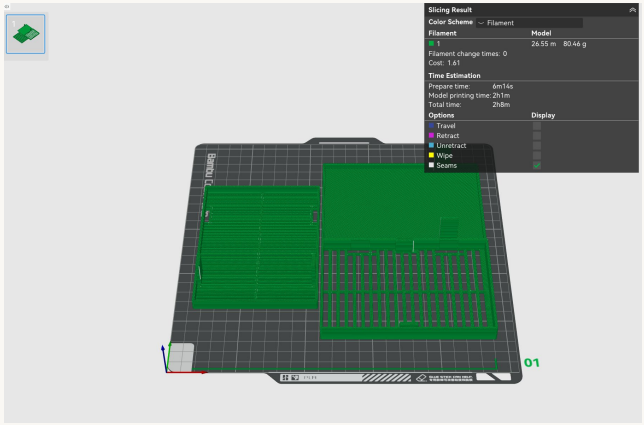
The photo shows one printed tray on the DeWalt. The sim shows the full stack it belongs in.

Prototype in action

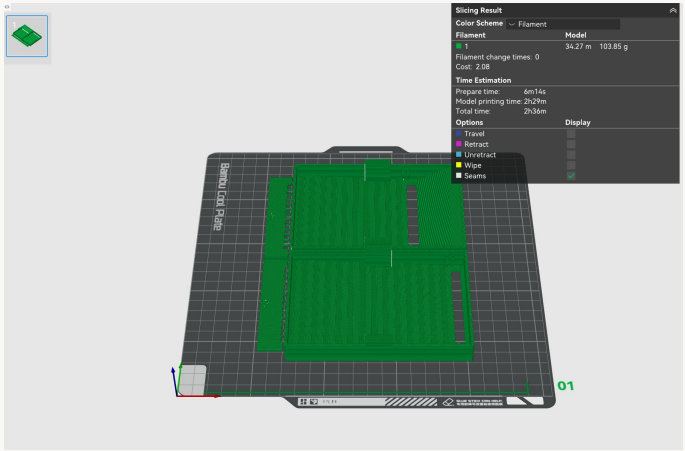


MuJoCo sim, uncalibrated (0.4x speed)





Plinko tower: 2 h 8 min, 80.5 g PLA



Gated tower: 2 h 36 min, 103.9 g PLA

Picture: Bench Prototype
First clip: Bench Prototype in action
Second clip: a 20 Hz, ±1.24 mm vertical stroke (assumed). Pasta stays on 5 mm, quinoa falls to the catch, and rice leaks through the 2 mm tray (see round 1).

RECOMMENDED SHAKERS

Shakers that can reproduce the motion we found (claude-based data)

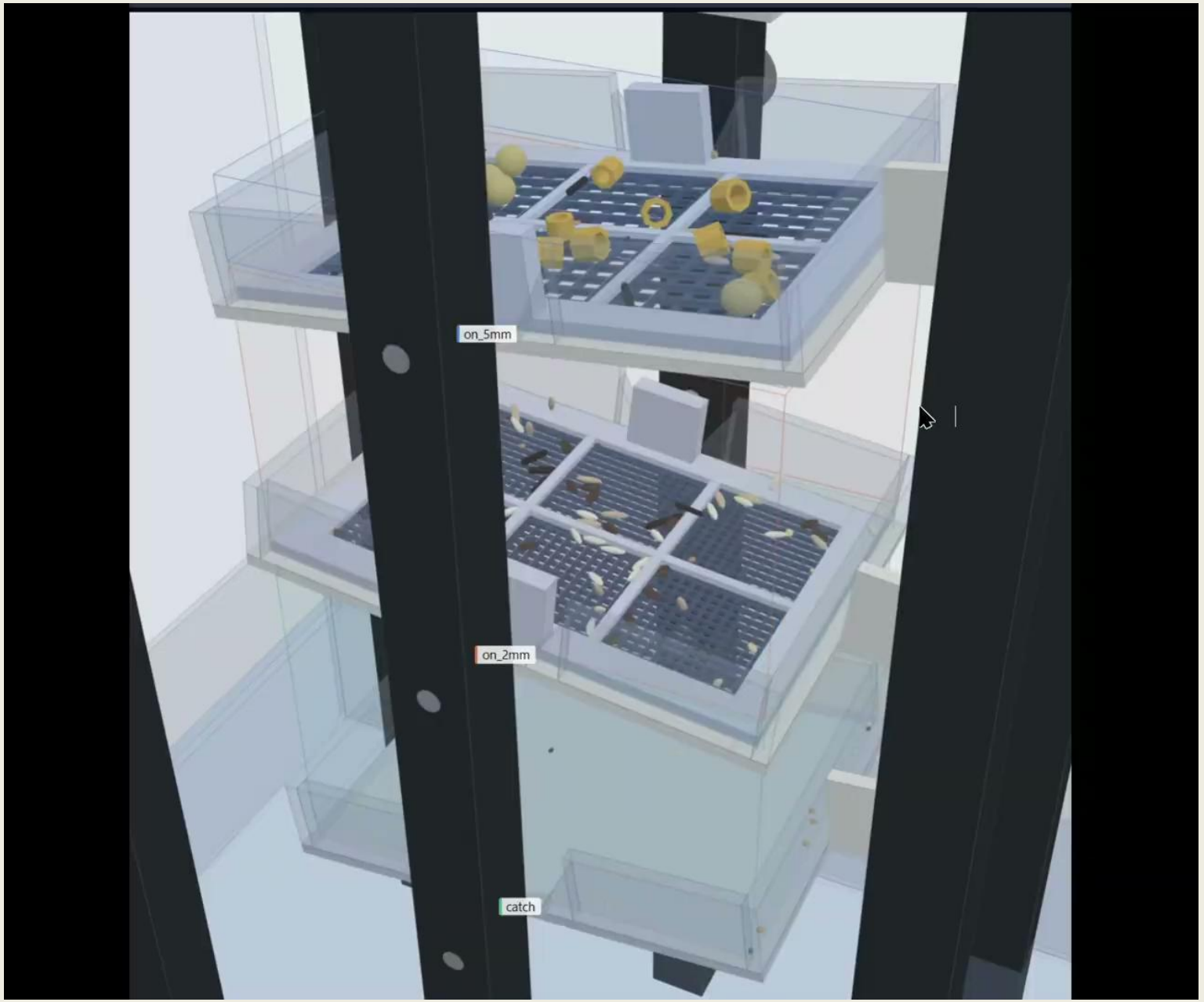
Shaker (brand · model)	Motion	Key specs	Wet sieving	Price
Retsch · AS 200 basic	3D throwing motion, electromagnetic	Amplitude 0–3 mm (digital); 200 mm / 8" sieves; up to 17 fractions; 4 kg stack; 35 kg	Yes	£3,113 ex VAT
W.S. Tyler · RO-TAP RX-29	Orbital + vertical tapping	278 osc/min, 150 taps/min; 6 full or 13 half 8" sieves; 82 kg	Dry	Quote
Gilson · SS-15 Economy	Lateral + up/down + tilt orbit	1/4 hp motor; 6 full or 13 half 8" sieves; 15-min timer	Dry	$3,037 list
Endecotts · Minor 200	Electromagnetic vibration	3,000/min at 50 Hz; ~1.6 mm fixed amplitude; 8 full 8" sieves; ~16 kg	No	Quote
Electric Vibrating Sieve Shaker Machine		1400r/Min 125w 110v with 200 Mesh Screens, 7-Layer Stainless Steel Lab Sifter, Sieves from 0.025-20mm	Yes (sealed)	$341 + shipping

Highlighted rows are our picks: the Retsch as the lab-grade benchmark, the Dayton array as the open, budget build.

Prototype 2 (only Sim)

Try the Simulator yourself!

- 8,200+ physics simulations
- 444 sieve designs scored automatically,
- 5 Design Types
- 8 particle types
  - Pasta, Rice, Grains



Thank You

Demo videos and files (Drive)

THE PROBLEM

A Malaise sample arrives as a clump

- **Thousands of tiny flies in ethanol,** tangled into a ball and mixed with leaves, twigs and dirt.
- **Step 1: break it apart, gently.** Every specimen goes on to imaging and DNA barcoding, so a crushed body or a snapped leg is a failure.
- **Step 2: sort by size** into BugPicker's classes (> 6 mm, 2–6 mm, < 2 mm). Leaves stay on top and dirt goes to the bottom.
- **Bench stand-ins:** pasta, rice and quinoa play the three size classes.



Bench test: our printed 5 mm tray, sorting pasta from rice

NEXT

Next: run the loop on real Malaise samples

- **Print the hole gauge** and correct the CAD to the printer's real opening sizes.
- **Print 1.4 to 1.8 mm gate strips for the rice cut** and bench-test them on the stack moving up and down.
- **Simulate the rig's real stroke** and calibrate the sim with the bench counts, so it predicts before we print.
- **Then insects:** a breaker layer on top to pull the Malaise clump apart, and gate sizes picked from specimen measurements.

HOW IT WORKS

Print, test, repeat: how we find the right size

Each round changes one size, so we learn what that size does.

1

Spec the sample

What's in it and how big: length, width, thickness. Which class should end on which tray.

2

Pick sizes

A size calculator proposes a mesh or gate per tray. The MuJoCo sim runs it first, with the stack moving up and down.

3

Generate and slice

CadQuery builds the part at that size. Bambu Studio gives the print time and filament: a gate strip is about 6 min.

4

Print and bench-test

Run pasta, rice and quinoa (then insects) through the stack. Count what ends up on each tray.

5

Log and adjust

Write the counts to the run log, calibrate the sim against them, and change one size.

Repeat until every tray holds one size class

A Claude Code skill in the repo (sieve-gate-design) walks through these steps and keeps the log, so anyone on the team can run the next round.